Diagnostics: residuals, QQ, leverage, Cook's distance, VIF

Inference labs use the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Read the four default `plot(lm)` diagnostics and say what each is for.
- Identify leverage and influence separately and combine them using Cook's distance.
- Detect collinearity with VIF and decide when to act on it.

Prerequisites

Sessions 2 and 3 of this week.

Background

Diagnostics are the step in a regression workflow that protects the report from the dataset. A coefficient is only as trustworthy as the assumptions behind it. The four classical plots — residuals vs fitted, QQ, scale-location, and residuals vs leverage — answer four questions: is the mean right, are the errors roughly normal, is the variance constant, and is any single point running the show?

Leverage measures how unusual a point's predictor values are; influence measures how much the fit changes when that point is removed. Cook's distance combines the two. A point can have high leverage without high influence if it sits on the regression line, and it can have high influence without spectacular residuals if its predictors are extreme.

The `performance::check_model()` function wraps many of these ideas in a single call and is increasingly the way teams read diagnostics on screen. `car::vif()` gives the variance inflation factors; a rule of thumb is that values above 5 deserve attention and above 10 demand it, but rules of thumb are no substitute for thinking about which predictors are truly redundant.

Setup

```r
library(tidyverse)
library(broom)
library(car)
library(performance)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

Simulate a regression with mild heteroscedasticity, a high-leverage point, and two correlated predictors. Ask which diagnostics flag which problem.

2. Visualise

```r
x1 <- rnorm(n, 0, 1)
x2 <- x1 + rnorm(n, 0, 0.3)       # x1 and x2 correlated
```

```r
x3 <- rnorm(n, 0, 1)
y  <- 1 + 2 * x1 - 1.5 * x2 + 0.5 * x3 + rnorm(n, 0, 1 + 0.4 * abs(x1))
# add one influential point
x1[1] <- 5; x2[1] <- 5; y[1] <- 0
dat <- tibble(y, x1, x2, x3)

ggplot(dat, aes(x1, y)) +
  geom_point(alpha = 0.6) +
  labs(x = "x1", y = "y")
```

3. Assumptions

```r
par(mfrow = c(2, 2))
plot(fit)
par(mfrow = c(1, 1))
```

The residuals-vs-fitted plot should show no trend. The QQ plot should track the line. Scale-location should be flat. Residuals-vs-leverage should have no points outside the Cook's-distance contours.

4. Conduct

Influence diagnostics:

```r
infl |> arrange(desc(.cooksd)) |> head(5) |>
  select(row, .fitted, .resid, .hat, .cooksd)
```

VIF:

```r
```

Performance summary:

```r
check_model(fit, check = c("vif", "qq", "outliers", "linearity"))
```

VIF for x1 and x2 should be high; x3 should be fine. The first row (the injected leverage point) should dominate Cook's distance.

5. Concluding statement

> Regression diagnostics identified one high-influence observation (Cook's $D =$ `round(max(infl$.cooksd), 2)`) and collinearity between x1 and x2 (VIF = `round(max(vif(fit)), 1)`). Refitting without the influential point or after combining the collinear predictors would be the next step before reporting coefficients.

Common pitfalls

- Treating a single Cook's distance number as a verdict. Always look at the plot.
- Solving a VIF problem by dropping one variable at a time when the underlying issue is that two predictors measure the same thing.
- Declaring the assumptions met because the numbers pass. Plot first.

Further reading

- Fox J, Weisberg S. *An R Companion to Applied Regression*, ch. 8.
- Belsley DA, Kuh E, Welsch RE. *Regression Diagnostics*.
- Lüdecke D et al. (2021), *performance: An R Package for Assessment...*

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)